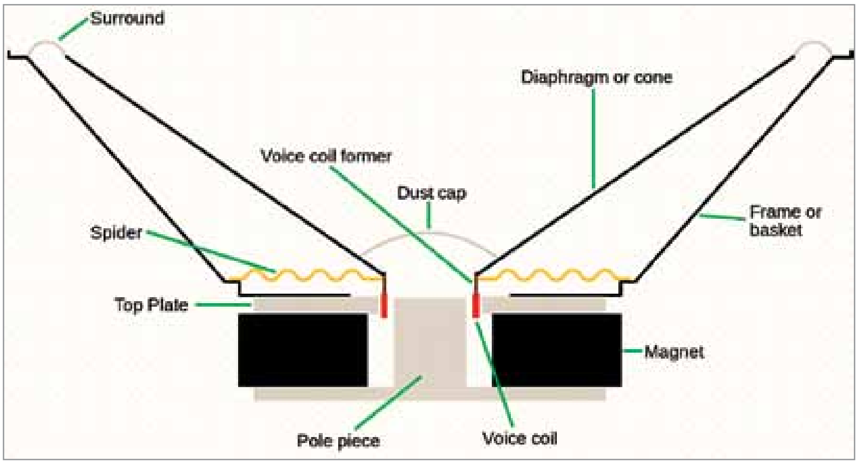


Fig. 11: Sectional arrangement of a woofer

wasted as heat, thereby increasing temperature of the voice coil. Coil's resistance increases with temperature @ 0.4% per degree centigrade, which can significantly degrade its performance. So, special techniques are used for making the coils.

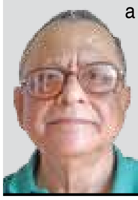
The best way to make a high-performance coil is to wind it on an alloy 1145 aluminium former by wet winding, while drawing the wire through a bath of high temperature adhesive comprising glass fibres impregnated with polyamide resin. This treatment makes the voice coil to survive up to a temperature of 300 degrees centigrade. Large magnet area, pole venting, large diameter of voice coil, and blackening of voice coil are some of the measures used to remove heat generated in the voice coil.

The performance of the speaker system can be further enhanced if magnetic fluid is used. In a conventional speaker, a voice coil is suspended on a damper made of cloth, called spider. As the voice coil moves up and down, the spider also vibrates. This generates secondary sound pressure waves that can cause distortion of sound. Solution to this problem is elimination of spider altogether and dipping the voice coil in micron-deep pool of ferrofluid.

Ferrofluid is a stable colloidal suspension of nano particles of magnetite in oil. The ferrofluid provides a free-flowing movement of the diaphragm to accurately reproduce the sound. The lack of dampening also means three decibels increase in volume and decrease of energy use by 35%.

Ferrofluid has approximately four times the heat transfer capacity of air, so it cools the voice coil very effectively. When applied to an eccentric voice coil, it distributes itself evenly around the air gap, forcibly centring the coil assembly, so there is no rub and scrape. A small amount of ferrofluid placed in the gap between the speaker magnet and voice coil is held in place by its attraction to the magnet and forms an excellent seal. It can increase short-time power handling up to 40 times and can be very useful for reproduction of classical music where short-time peak power requirement is often 100 times more than the average power.

Ferrofluid is not generally used in woofers, because their high excursion would tend to throw the fluid out of the voice coil gap. Ferrofluid can be very effectively used in tweeter as it has low wattage and can reduce clipping of sound to a great extent. As there is very little friction, it reduces energy consumption by 35% for reproducing the same loudness of sound as conventional speakers. It also improves the sound quality by nearly 3dB. This type of tweeter with ferrofluid, if used, would further enhance the performance of spiral speakers. EFY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